Electronic Appliance Store Management System

Database Management Systems (DBMS) — Review 2: Database Implementation & Backend Integration

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Subject	Database Management Systems
Evaluation	Review 2 — Database Implementation & Backend Integration
Database	Oracle Database 21c XE, pluggable database `XEPDB1`, schema `appliance_app`
Backend	Python Flask, python-oracledb (Thin mode)
Frontend	HTML5, CSS3, vanilla JavaScript
Environment	Windows local demonstration

> **Verification disclosure.** This report was produced in an environment with no Oracle Database. Every statement about Oracle behaviour is therefore *designed and statically checked, not live-verified*. Live Oracle tests are labelled **BLOCKED** in the testing matrix (§28). They must be executed locally (README §9) before the live-database criteria are claimed as verified.

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1. Abstract

The Electronic Appliance Store Management System is a database-driven web application for a retail appliance shop. A normalized Oracle 21c schema of seven tables stores categories, suppliers, products, customers, employees, sales and sale items. A Flask service layer using python-oracledb provides CRUD for four master entities, transactional multi-item sales with row-level inventory locking, a dashboard and five reports. Oracle-side logic consists of one view, one stored procedure, one stored function and one integrity trigger. Sixteen documented SQL queries demonstrate joins, subqueries, aggregation and analytic features. Security controls include bind variables, CSRF tokens, escaped output, a least-privilege database account and sanitized errors. 76 automated tests ran and passed without Oracle (fake connection); 16 live-Oracle tests are written and BLOCKED pending a local Oracle instance.

2. Introduction

Appliance retailers handle high-value items with changing prices and limited stock. Spreadsheet records lead to oversold items, lost price history and inconsistent customer data. A relational database with enforced integrity and transactional sales processing addresses these issues. This project implements that system for Review 2, which evaluates database creation, keys and constraints, sample data, queries, advanced objects, connectivity and CRUD/transactions.

3. Problem statement

Design and implement a relational database and web front end that (a) stores store master data without redundancy, (b) records sales with historical prices, (c) prevents negative or oversold inventory even with simultaneous users, and (d) answers business questions (revenue, best sellers, low stock, customer spend).

4. Objectives

• Create a normalized (3NF) Oracle schema with full integrity constraints.

• Provide deterministic fictional sample data.

• Provide exactly 16 documented queries and four advanced database objects.

• Connect Flask to Oracle securely with python-oracledb Thin mode.

• Implement CRUD and an atomic sales transaction that never oversells.

• Provide automated tests with honest, status-labelled results.

5. Scope and limitations

In scope: master-data CRUD, sales entry/history, inventory reports, restock. Out of scope: user authentication/authorization, returns and refunds, taxes/discounts, purchase orders to suppliers, multi-currency, employee CRUD screens (employees are seeded and selectable on sales). Further limitations in §26.

6. Functional requirements

• FR1 Manage products, categories, suppliers, customers (create, list/search, edit, delete).

• FR2 Record a sale with one or more products; totals and prices come from the database.

• FR3 Deduct stock exactly once per sale; reject insufficient stock.

• FR4 Restock a product.

• FR5 Browse sales history and sale details; search by invoice, customer or employee.

• FR6 Dashboard (counts, revenue, low-stock count, recent sales).

• FR7 Reports: inventory status, revenue by month, top products, revenue by category, customer purchase summary.

• FR8 Health endpoint reporting real database connectivity.

7. Non-functional requirements

• Integrity: all rules enforced in the database where possible (PK/FK/UNIQUE/CHECK).

• Security: no secrets in code, bind variables, CSRF, output escaping, least privilege.

• Reliability: transactions commit atomically or roll back; connections always closed.

• Usability: responsive layout, empty/error/success states, delete confirmation.

• Maintainability: service-layer separation; tests per layer.

• Portability: Windows 10/11 local setup.

8. Software and hardware requirements

Item	Requirement
OS	Windows 10/11 (development sandbox was Linux, used only for non-Oracle checks)
Python	3.11+ (3.13 used in the sandbox)
Python packages	Flask, oracledb, python-dotenv, pytest (python-docx, reportlab for report build)
Database	Oracle Database 21c XE with XEPDB1
Tools	SQL*Plus (bundled with XE), PowerShell, a modern browser
Hardware	2+ CPU cores, 4 GB RAM minimum (Oracle XE limit 2 GB database RAM), ~5 GB disk

9. System architecture

```
Browser (HTML/CSS/JS)
   ■  HTTP (forms, CSRF token)
   ▼
Flask routes ■■ app.py (validation of route args, CSRF, error pages, security headers)
   ■
   ▼
Service layer ■■ services/{products,categories,suppliers,customers,employees,sales,reports}.py
   ■              services/validation.py  (server-side input rules)
   ■              services/database.py    (connection, bind-variable helpers, error translation)
   ▼
python-oracledb (Thin) ■■ config.py (env vars only)
   ▼
Oracle 21c XE / XEPDB1 / schema appliance_app
   tables · indexes · identity columns · sequence · view · procedure · function · trigger
```

Routes call services only; services own all SQL; the database layer is the only module importing `oracledb`'s connect call.

10. Database design

Naming: upper-case Oracle identifiers; `PK_`, `FK_`, `UQ_`, `CK_`, `IX_` prefixes. Surrogate keys use `NUMBER GENERATED ALWAYS AS IDENTITY`. Money columns are `NUMBER(p,2)` (Python `Decimal`, never float). Timestamps use `TIMESTAMP`. Deletion behaviour: every foreign key is `NO ACTION` (restrict) — there is no cascade, so historical sales cannot be destroyed by deleting a customer, employee or product (ORA-02292, reported to users as a friendly 409). Indexes: FK columns (`IX_PRODUCTS_CATEGORY`, `IX_PRODUCTS_SUPPLIER`, `IX_SALES_CUSTOMER`, `IX_SALES_EMPLOYEE`, `IX_SALE_ITEMS_PRODUCT`) and the report filter `IX_SALES_DATE`; PK/UNIQUE constraints supply their own indexes (including `UQ_SALE_ITEMS_LINE (SALE_ID, PRODUCT_ID)` which serves lookups by sale).

11. ER diagram

```
erDiagram
    CATEGORIES ||--o{ PRODUCTS : classifies
    SUPPLIERS  ||--o{ PRODUCTS : supplies
    CUSTOMERS  ||--o{ SALES : places
    EMPLOYEES  ||--o{ SALES : processes
    SALES      ||--|{ SALE_ITEMS : contains
    PRODUCTS   ||--o{ SALE_ITEMS : "appears in"
    CATEGORIES { number CATEGORY_ID PK
                 varchar2 CATEGORY_NAME UK
                 varchar2 DESCRIPTION }
    SUPPLIERS  { number SUPPLIER_ID PK
                 varchar2 SUPPLIER_NAME UK
                 varchar2 CONTACT_PERSON
                 varchar2 EMAIL
                 varchar2 PHONE
                 varchar2 ADDRESS }
    PRODUCTS   { number PRODUCT_ID PK
                 varchar2 PRODUCT_NAME
                 varchar2 BRAND
                 varchar2 MODEL_NO UK
                 number CATEGORY_ID FK
                 number SUPPLIER_ID FK
                 number UNIT_PRICE
                 number STOCK_QTY
                 number REORDER_LEVEL
                 number WARRANTY_MONTHS
                 timestamp CREATED_AT }
    CUSTOMERS  { number CUSTOMER_ID PK
                 varchar2 FULL_NAME
                 varchar2 EMAIL UK
                 varchar2 PHONE
                 varchar2 ADDRESS
                 timestamp CREATED_AT }
    EMPLOYEES  { number EMPLOYEE_ID PK
                 varchar2 FULL_NAME
                 varchar2 EMAIL UK
                 varchar2 PHONE
                 varchar2 JOB_TITLE
                 date HIRE_DATE }
    SALES      { number SALE_ID PK
                 varchar2 INVOICE_NO UK
                 number CUSTOMER_ID FK
                 number EMPLOYEE_ID FK
                 timestamp SALE_DATE
                 varchar2 PAYMENT_METHOD
                 number TOTAL_AMOUNT }
    SALE_ITEMS { number SALE_ITEM_ID PK
                 number SALE_ID FK
                 number PRODUCT_ID FK
                 number QUANTITY
                 number UNIT_PRICE
                 number LINE_TOTAL }
```

Text alternative (crow's foot: `1` = one, `N` = many):

```
CATEGORIES 1■■■N PRODUCTS N■■■1 SUPPLIERS
                  ■1
                  N
CUSTOMERS 1■■■N  SALE_ITEMS N■■■1 SALES N■■■1 EMPLOYEES
                                ▲N
                                ■■■■■■■1 CUSTOMERS
(SALES 1■■■N SALE_ITEMS ; PRODUCTS 1■■■N SALE_ITEMS ; CUSTOMERS 1■■■N SALES ; EMPLOYEES 1■■■N SALES)
```

12. Table descriptions and relationships

Table	Purpose	Key columns
CATEGORIES	Product groups	PK `CATEGORY_ID`; UK `CATEGORY_NAME`
SUPPLIERS	Vendors	PK `SUPPLIER_ID`; UK `SUPPLIER_NAME`
PRODUCTS	Central product catalogue and live stock	PK `PRODUCT_ID`; UK `MODEL_NO`; FK `CATEGORY_ID`, `SUPPLIER_ID`
CUSTOMERS	Buyers	PK `CUSTOMER_ID`; UK `EMAIL`
EMPLOYEES	Staff who process sales	PK `EMPLOYEE_ID`; UK `EMAIL`
SALES	One row per invoice	PK `SALE_ID`; UK `INVOICE_NO`; FK `CUSTOMER_ID`, `EMPLOYEE_ID`
SALE_ITEMS	Lines of a sale with historical price/quantity	PK `SALE_ITEM_ID`; UK (`SALE_ID`,`PRODUCT_ID`); FK `SALE_ID`, `PRODUCT_ID`

Relationships (all one-to-many): a category has many products; a supplier supplies many products; a customer has many sales; an employee processes many sales; a sale contains one or more items (≥1 enforced by the service); a product appears in many sale items. A product row is never duplicated by a sale — SALE_ITEMS references it and keeps the price and quantity at the time of sale.

13. Primary keys, foreign keys and integrity constraints

Table	Constraints (names as in schema.sql)
CATEGORIES	PK_CATEGORIES, UQ_CATEGORIES_NAME, CATEGORY_NAME NOT NULL
SUPPLIERS	PK_SUPPLIERS, UQ_SUPPLIERS_NAME, SUPPLIER_NAME NOT NULL
CUSTOMERS	PK_CUSTOMERS, UQ_CUSTOMERS_EMAIL, FULL_NAME/EMAIL/PHONE NOT NULL
EMPLOYEES	PK_EMPLOYEES, UQ_EMPLOYEES_EMAIL, FULL_NAME/EMAIL/JOB_TITLE/HIRE_DATE NOT NULL
PRODUCTS	PK_PRODUCTS, UQ_PRODUCTS_MODEL, FK_PRODUCTS_CATEGORY, FK_PRODUCTS_SUPPLIER, CK_PRODUCTS_PRICE (UNIT_PRICE >= 0), CK_PRODUCTS_STOCK (STOCK_QTY >= 0), CK_PRODUCTS_REORDER (>= 0), CK_PRODUCTS_WARRANTY (0..120)
SALES	PK_SALES, UQ_SALES_INVOICE, FK_SALES_CUSTOMER, FK_SALES_EMPLOYEE, CK_SALES_PAYMENT (CASH, CARD, UPI, BANK_TRANSFER), CK_SALES_TOTAL (TOTAL_AMOUNT > 0)
SALE_ITEMS	PK_SALE_ITEMS, UQ_SALE_ITEMS_LINE (SALE_ID, PRODUCT_ID), FK_SALE_ITEMS_SALE, FK_SALE_ITEMS_PRODUCT, CK_SALE_ITEMS_QTY (> 0), CK_SALE_ITEMS_PRICE (>= 0), CK_SALE_ITEMS_TOTAL (LINE_TOTAL = QUANTITY * UNIT_PRICE)

E-mail and phone *format* validation is performed at the application level (services/validation.py) as specified. Sequence SEQ_INVOICE_NO (start 1000) generates invoice numbers INV-001000…; seed invoices use INV-000001… so they never collide.

14. Normalization analysis (1NF → 3NF)

1NF. Every column holds a single atomic value and no table has repeating groups: the "many products in one sale" requirement is modelled by rows in SALE_ITEMS rather than columns such as PRODUCT1, PRODUCT2. Each table has a primary key. (An address is kept as one atomic text attribute because the application never queries its parts; splitting it would be a refinement, not a 1NF requirement here.)

2NF. All tables use a single-column surrogate primary key, so partial dependencies on part of a key cannot occur. For SALE_ITEMS the natural candidate key is the composite (SALE_ID, PRODUCT_ID): QUANTITY and UNIT_PRICE depend on *both* (how many of which product on which invoice). Product attributes such as name or category are deliberately **not** copied into SALE_ITEMS; they would depend on PRODUCT_ID alone (a partial dependency) and are kept only in PRODUCTS.

3NF. No non-key attribute depends on another non-key attribute, with the examples:

Table	Functional dependencies	Why in 3NF
CATEGORIES	CATEGORY_ID → CATEGORY_NAME, DESCRIPTION; CATEGORY_NAME → CATEGORY_ID	Both determinants are candidate keys
SUPPLIERS	SUPPLIER_ID → all; SUPPLIER_NAME → all	Determinants are keys
PRODUCTS	PRODUCT_ID → all; MODEL_NO → all	Category and supplier *names* are not stored — only FKs — so no PRODUCT_ID → CATEGORY_ID → CATEGORY_NAME transitive dependency
CUSTOMERS	CUSTOMER_ID → all; EMAIL → all	Determinants are keys
EMPLOYEES	EMPLOYEE_ID → all; EMAIL → all	Determinants are keys
SALES	SALE_ID → all; INVOICE_NO → all	Customer/employee names are not stored (only FKs)

Table	Functional dependencies	Why in 3NF
SALE_ITEMS	`SALE_ITEM_ID` → all; (`SALE_ID`, `PRODUCT_ID`) → `QUANTITY`, `UNIT_PRICE`, `LINE_TOTAL`	See the documented exception below

Documented, deliberate redundancy (not hidden).

• `SALE_ITEMS.LINE_TOTAL = QUANTITY × UNIT_PRICE` is a derived column: {`QUANTITY`, `UNIT_PRICE`} → `LINE_TOTAL` has a non-key determinant, so strictly `SALE_ITEMS` is *not* in 3NF. It is stored for audit and fast reporting, and the anomaly risk is removed by `CK_SALE_ITEMS_TOTAL`.

• `SALES.TOTAL_AMOUNT` equals the sum of its lines' `LINE_TOTAL`. It is computed once by the service from database prices; `V_SALES_SUMMARY.TOTAL_MATCHES` exposes any mismatch. A CHECK cannot span tables, so this one is enforced by the service, not by a constraint.

• `SALE_ITEMS.UNIT_PRICE` repeats the *name* of `PRODUCTS.UNIT_PRICE` but not its meaning: it is the price at the moment of sale (history), whereas the product price is the current price.

15. Implementation details

• `database/schema.sql` — re-runnable DDL via an anonymous block that tolerates only ORA-00955.

• `database/seed.sql` — generated from fixed data; `MERGE` by natural keys; totals computed, not hand-typed.

• `database/queries.sql` — 16 numbered queries; `database/advanced_features.sql` — four objects plus self-rolling-back verification blocks; `database/setup_user.sql` — one-time admin script (password typed at a hidden prompt).

• Python: `config.py`; `services/*`; `app.py` (25 routes); Jinja templates and static assets.

16. Flask and Oracle connectivity

`config.load_db_settings()` reads `DB_HOST, DB_PORT, DB_SERVICE_NAME, DB_USER, DB_PASSWORD` from the environment (`python-dotenv` loads `.env` at app start). Values still in `<placeholder>` form count as missing; errors list variable *names* only; `SYS/SYSTEM/SYSDBA/SYSOPER` are refused. `services.database.get_connection()` opens a Thin-mode connection per unit of work (`oracledb.connect(user, password, dsn="host:port/XEPDB1")`) and `connection()` always closes it (rolling back on error). `oracledb.defaults.fetch_decimals = True` returns money as `Decimal`. Diagnostics: `python -m services.database` and `GET /health` — both report success only after `SELECT USER … FROM DUAL` returned. Unavailable Oracle yields HTTP 503 with remediation text and no invented figures.

17. CRUD operations

Products, categories, suppliers and customers each have list (search/filter/pagination), create, update (edit form pre-filled via `?edit=<id>`) and delete (POST, confirmation dialog, CSRF). Each service validates input, then runs a parameterized statement in its own short transaction. Constraint errors are translated: ORA-00001 → 409 "already exists", ORA-02291 → 400 invalid reference, ORA-02292 → 409 "referenced by other records", ORA-02290 → 400 rule violation, ORA-30006 → 409 "inventory busy". Restock calls `SP_RESTOCK_PRODUCT`.

18. Transaction management and inventory consistency

`services/sales.create_sale` (single connection, no autocommit):

• Validate ids, payment method and items (non-empty, quantity 1–1000, ≤50 lines; duplicate products merged). No SQL runs if validation fails.

• Verify the customer and employee exist.

• For each product **in ascending id order** run `SELECT … FOR UPDATE WAIT 10` (consistent lock order → no deadlock between sales; bounded wait → ORA-30006 instead of hanging).

• After *all* locks are held, re-check stock; any shortage → `InsufficientStockError` listing the products.

• Compute each line and the total as `Decimal` from the **locked database rows** (client prices/totals are never accepted — `create_sale` has no price/total parameters).

• `INSERT` the sale (invoice from `SEQ_INVOICE_NO`, `RETURNING SALE_ID`), `INSERT` each item, `UPDATE PRODUCTS SET STOCK_QTY = STOCK_QTY - :qty` — the **only** place stock is deducted (the trigger deliberately does not touch inventory).

• One `COMMIT`. Any exception → rollback (the `connection()` context manager and the connection close both enforce it).

Oracle's default `READ COMMITTED` isolation plus row locks prevents overselling: a concurrent buyer blocks on the lock, then re-reads the committed lower stock. `CK_PRODUCTS_STOCK` is the final safety net. Unit tests prove ordering, single commit, rollback and parameterization against a fake connection; the live concurrency proof is integration test INT "concurrent sales cannot oversell" (**BLOCKED** here).

19. SQL joins, subqueries and aggregations

#	Query	Concept
1	In-stock products over ■20,000	SELECT, WHERE, ORDER BY
2	Products in categories matching a pattern	IN subquery
3	Products with category	INNER JOIN
4	Products with supplier	JOIN
5	Customers with their sales (incl. none)	LEFT JOIN
6	Employee sales performance	LEFT JOIN + COUNT/SUM
7	Full invoice lines	5-table JOIN
8	Products per category	GROUP BY + COUNT
9	Monthly revenue	SUM/COUNT/AVG by date expression
10	Categories with revenue > 50,000	HAVING
11	Products above average price	Scalar subquery
12	Products above their category average	Correlated subquery
13	Products never sold	NOT EXISTS
14	Top 5 products by units	Aggregation + FETCH FIRST
15	Customer spend ranking	RANK() analytic
16	Inventory value and reorder status	CASE KPI

Status: all 16 statements parse with `sqlglot` (Oracle dialect) — a static syntax check only; **live execution on Oracle is BLOCKED**. With the seed data Q13 should return `VV-TV32` and `AU-EK18`, Q10 should list Televisions (■139,996), Air Conditioners (■63,498) and Washing Machines (■50,998), Q16 flags `CB-AC10` ($2 \le 3$) and `VV-TV55` ($4 \le 4$) as REORDER NOW (expected values derived by hand from the seed; not yet confirmed on Oracle).

20. View, procedure, function and trigger

Object	Purpose	Errors / behaviour
View `V_SALES_SUMMARY`	One row per sale with customer, employee, line/unit counts and `TOTAL_MATCHES`	Sales without items still appear (LEFT JOIN); used by sales history and dashboard
Procedure `SP_RESTOCK_PRODUCT(p_product_id, p_quantity)`	Add stock under a row lock; does not commit (caller owns the transaction)	ORA-20001 invalid quantity, ORA-20002 unknown product; used by the Restock button
Function `FN_CUSTOMER_TOTAL_SPEND(p_customer_id)`	Lifetime spend of a customer, 0 if none	ORA-20010 unknown customer; callable from SQL
Trigger `TRG_SALES_PROTECT_HISTORY`	Integrity: blocks DELETE of a sale and UPDATE of customer, employee, invoice, date, total, payment	ORA-20030 / ORA-20031; no inventory logic; documented test-cleanup bypass via `CLIENT_INFO`

Each has example invocation and verification SQL in `advanced_features.sql` (the verification blocks roll back). Live execution: **BLOCKED**.

21. Frontend implementation

Server-rendered Jinja templates with a shared `base.html` (navigation, flash messages), responsive CSS (grid, `prefers-color-scheme`, mobile breakpoint, visible focus), and ~60 lines of vanilla JavaScript: delete confirmation (`data-confirm`), dynamic add/remove of sale lines, and a clearly labelled *estimated* total. Every button and form maps to a real route; pagination links preserve filters; empty, error (503/409/400) and success states exist; the page works without JavaScript except the extra sale-line rows. No framework or external dependency; Content-Security-Policy `default-src 'self'` forbids inline scripts.

22. Security controls

Control	Implementation
SQL injection	Bind variables only; the only formatted SQL fragments are developer constants (column lists, sorted ORDER BY, the integer lock-wait)
CSRF	Per-session token required on every POST/PUT/PATCH/DELETE (form field or `X-CSRF-Token`), constant-time compare
XSS	Jinja auto-escaping; no `innerHTML`; CSP
Secrets	`.env` git-ignored; `.env.example` placeholders; errors/logs never contain passwords; `repr` of settings hides the password
Least privilege	`appliance_app` has six `CREATE …` privileges; config refuses admin users; admin setup is a separate manual script
Sessions	HttpOnly, SameSite=Lax, Secure in production
Errors	Safe messages, ORA text only in server log
Integrity	CHECK constraints reject negative price/stock/quantity; trigger protects history
Debug	Off unless `APP_ENV` ≠ production **and** `FLASK_DEBUG=1`

Not production-secure: there is **no authentication or authorization**, no rate limiting, no audit log of users, HTTPS is not provided by the development server.

23. Testing methodology

pytest suites: `test_config` (settings), `test_services` (validation, CRUD and transaction logic against a fake connection that records SQL, parameters, commits, rollbacks and closes), `test_routes` (Flask test client with stubbed services: CSRF, status codes, escaping, headers, health), `test_oracle_integration` (live Oracle, separate test schema, tagged rows, explicit opt-in). Status vocabulary — **PASSED**: executed and passed · **FAILED**: executed and failed · **NOT RUN**: intentionally skipped · **BLOCKED**: required dependency unavailable. Fakes prove application logic, not Oracle semantics (locking, constraint enforcement, PL/SQL), which is why the integration suite exists. Mutation sanity checks: removing the lock-order sort, and adding a mid-sale commit, each made tests fail (then reverted).

24. Verified testing matrix

Generated from the actual pytest result file (`docs/test_results.xml`) by `docs/build_reports.py`.

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
CFG-01	Valid DB_* variables produce settings and the XEPDB1 DSN	host:port/service DSN, no error	Executed; assertions held	**PASSED**	Pure logic
CFG-02	Missing DB_PASSWORD and DB_HOST are reported by NAME only	ConfigError lists names, leaks no value	Executed; assertions held	**PASSED**	Pure logic
CFG-03	Values copied unchanged from .env.example (<set-locally>)	treated as missing configuration	Executed; assertions held	**PASSED**	Pure logic
CFG-04	DB_USER set to an administrator account (3 parameter sets)	ConfigError (app must use appliance_app only)	Executed; assertions held	**PASSED**	Pure logic

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
CFG-05	Non-numeric or out-of-range DB_PORT (3 parameter sets)	ConfigError	Executed; assertions held	**PASSED**	Pure logic
CFG-06	repr() of the settings object	does not contain the password	Executed; assertions held	**PASSED**	Pure logic
CFG-07	FLASK_SECRET_KEY missing or placeholder	ConfigError with generation hint	Executed; assertions held	**PASSED**	Pure logic
CFG-08	APP_ENV=production	secure cookies on, debug off even if FLASK_DEBUG=1	Executed; assertions held	**PASSED**	Pure logic
INT-01	Connect to XEPDB1 as the test application user	connected, user is not SYS/SYSTEM	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-02	USER_TABLES	all 7 required tables exist	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-03	USER_OBJECTS	view, procedure, function, trigger all VALID	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-04	Negative price/stock, duplicate model, bad category FK	ORA-02290 / ORA-00001 / ORA-02291	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-05	SALE_ITEMS with LINE_TOTAL != QUANTITY*UNIT_PRICE	ORA-02290	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-06	Delete category/customer/product that history references	ConflictError (ORA-02292 translated)	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-07	Create, read, update, delete a category and customer through the services	each step reflected in DB	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-08	SP_RESTOCK_PRODUCT	+5 stock on success; ORA-20001 for 0, ORA-20002 unknown product	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-09	FN_CUSTOMER_TOTAL_SPEND	0 before sales, equals sale total after, ORA-20010 for unknown customer	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
INT-10	TRG_SALES_PROTECT_HISTORY	DELETE -> ORA-20030, UPDATE total -> ORA-20031	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-11	V_SALES_SUMMARY	shows the new sale with correct line/unit counts and TOTAL_MATCHES = 'Y'	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-12	Two-item sale	totals from DB prices (2*100.00+3*49.50=348.50), stock 10->8 and 5->2, items stored	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-13	Quantity above stock	InsufficientStockError; no sale row, stock unchanged	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-14	Valid line + nonexistent product	ValidationError and the valid line's stock is untouched	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-15	Another session holds FOR UPDATE on a product	sale waits then fails with ConflictError (ORA-30006), stock intact	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
INT-16	8 threads each buy 1 unit of a product with 5 in stock	exactly 5 succeed, 3 InsufficientStockError, final stock 0	BLOCKED: live Oracle tests not enabled (set RUN_ORACLE_TESTS=1 and TEST_DB_* variables)	**BLOCKED**	Requires live Oracle test schema
RTE-01	Dashboard with stubbed DB figures	200 and the figures are rendered	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-02	Oracle unavailable	HTTP 503 error page, no statistics shown	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-03	Category name containing <script>	rendered escaped, never as live markup	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-04	POST with no CSRF token	400 and the service is never called	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-05	POST with a forged CSRF token	400	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-06	Valid CSRF token + valid form	service called, 303 redirect (post/redirect/get)	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-07	X-CSRF-Token header instead of form field	accepted	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-08	GET on delete / update / restock URLs	405 Method Not Allowed	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-09	Service raises ValidationError	HTTP 400 with the message	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-10	Delete of a referenced row	HTTP 409 with friendly message	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-11	Sale rejected for insufficient stock	HTTP 409 with product detail	Executed; assertions held	**PASSED**	Services stubbed; Flask test client

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
RTE-12	Forged price/total fields in the sale POST	never forwarded to create_sale	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-13	Sale form with no product lines	HTTP 400, create_sale never reached	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-14	Unknown URL -> 404; non-numeric sale id -> 404	Unknown URL -> 404; non-numeric sale id -> 404	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-15	Unexpected exception text with a secret	HTTP 500, page shows a generic message only	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-16	Any response	CSP, nosniff and frame protection headers set	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-17	No Oracle configured	/health returns 503 and database=unavailable, never 'ok'	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-18	health_check reporting success	/health 200 with no secrets in the body	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-19	Products, customers, sales, sale detail, new sale, reports pages	all render 200 with stubbed data	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
SVC-01	Valid monetary strings (3 parameter sets)	accepted and quantised to 2 decimals	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-02	Negative, non-numeric, 3-decimal, NaN, too-large prices (6 parameter sets)	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-03	Malformed email / phone numbers	ValidationError; valid ones accepted	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-04	Zero, negative, non-integer, None, absurd IDs (6 parameter sets)	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-05	Duplicate products merged, lock order ascending	[(1, 3), (5, 2)]	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-06	Blank form rows ignored; only blanks -> empty sale rejected	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-07	Zero, negative, non-numeric, fractional or huge quantity (5 parameter sets)	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-08	Two-item sale	one commit, DB-priced total 35597.00 (Decimal), stock deducted per item, connection closed	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
SVC-09	Items supplied as 3,1,2	FOR UPDATE issued for 1,2,3 (consistent order -> no deadlock), all before any INSERT	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-10	Request 3 of a product with 2 in stock	InsufficientStockError, nothing inserted or updated, no commit, rolled back	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-11	Product id 999 does not exist	ValidationError, no writes, no commit	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-12	Customer id missing in DB	ValidationError before any product is locked	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-13	Stock UPDATE fails (CHECK violation) after sale+item inserted	ValidationError, rollback, NO commit	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-14	Bad payment method / empty items / bad ids (4 parameter sets)	ValidationError and zero SQL executed	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-15	Every statement uses bind variables; create_sale has no price/total input	no literals from user data in SQL	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-16	advanced_features.sql trigger body	contains no UPDATE of PRODUCTS (single deduction point)	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-17	SQL-injection text in a product name	stays in bind params, never in the SQL string	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-18	Negative price or stock in the form	ValidationError before any SQL	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-19	UPDATE affecting 0 rows	NotFoundError (404)	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-20	ORA-02292 on delete	ConflictError with friendly text, no ORA code/constraint name leaked	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-21	ORA-00001 on insert	ConflictError (409) with generic message	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-22	Listener down (ORA-12541)	DatabaseUnavailableError (503) without host/user/password details	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
SVC-23	Restock 10 units	SP_RESTOCK_PRODUCT called with (id, qty), exactly one commit	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-24	Procedure raises ORA-20002	NotFoundError, rolled back	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-25	oracledb.connect raising ORA-01017 (bad login)	DatabaseUnavailableError, no credentials in message	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-26	No DB configured	health_check returns (False, message) - never claims success	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour

Totals by pytest case: BLOCKED 16, PASSED 76

Other checks executed (sandbox: Linux, Python 3.13, no Oracle)

Check	Command	Result
Python compilation	`python -m compileall -q app.py config.py services tests`	PASSED
JavaScript syntax	`node --check static/js/app.js`	PASSED
Query count	`grep -c "^-- Query [0-9]*:" database/queries.sql`	16 — PASSED
SQL static parse	`sqlglot` (Oracle dialect) over 16 queries and 7 seed MERGE statements	PASSED (syntax only; PL/SQL scripts not parsed — NOT RUN)
Flask smoke	`python app.py` then `GET /health`, `GET /products`, `POST /categories` without token	503, 503, 400 as designed — PASSED
Connection diagnostic	`python -m services.database` with an unreachable listener	exit 1, sanitized message, no secrets — PASSED
Secret scan	recursive grep for credential-like assignments	only obviously fake test dummies; PASSED
Live Oracle: schema, seed, queries, PL/SQL, integration tests	see README §7, §9	**BLOCKED** — no Oracle instance in the authoring environment

25. Results and discussion

Everything that can run without Oracle ran and passed: configuration rules, validation, service logic including the full sale transaction path (lock order, stock re-check, single commit, rollback on failure), error translation (no leakage of credentials or ORA internals), route behaviour (CSRF, 400/404/409/503 statuses, XSS escaping, security headers, honest health reporting). The design choice of one deduction point (service) and sorted row locks gives deterministic concurrency behaviour that is straightforward to test. What remains unproven until run on Oracle: syntax acceptance of the DDL/PL/SQL scripts, actual constraint and trigger behaviour, the expected query results quoted in §19, and the concurrency tests. These are exactly the BLOCKED rows above.

26. Known limitations

• No authentication/authorization or per-user audit trail.

• Not verified on a live Oracle instance by the authoring environment (BLOCKED items).

• `SALES.TOTAL_AMOUNT` ↔ line totals consistency is service-enforced, not a constraint; `LINE_TOTAL` is a documented derived column.

• Trigger contains a test-cleanup bypass that should be removed for production-style use.

• No returns, discounts, taxes, multi-currency; employees have no CRUD screen.

• `LIKE` searches treat %/_ as wildcards; per-request connections (no pool) are adequate for a demo, not for load.

• The seed is first-run oriented: it never overwrites changed rows, so edits to seeded products persist.

27. Installation and execution summary

See README §4–§9. In brief (PowerShell): create `appliance_app` once as admin with `database\setup_user.sql`; `py -3.11 -m venv .venv`; activate; `pip install -r requirements.txt`; copy `.env.example` to `.env` and fill in the password and a generated secret key; run `schema.sql`, `advanced_features.sql`, `seed.sql` with `sqlplus appliance_app@localhost:1521/XEPDB1 @database\<file>.sql`; `python -m services.database`; `python -m pytest tests -v`; `python app.py`; open `http://127.0.0.1:5000`.

28. Faculty evaluation cross-reference

#	Criterion	Where implemented	How to demonstrate	Verification status
1	Database and table creation	`database/schema.sql`, `setup_user.sql`, README §5–§7	Run scripts; `SELECT table_name FROM user_tables`	Scripts written; live run **BLOCKED**
2	Keys and constraints	PK/FK/UQ/NOT NULL/CHECK in `schema.sql`; §13	`user_constraints` query; demo_guide §2 failing inserts	Live enforcement **BLOCKED** (integration tests INT-04/05/06 exist)
3	Meaningful sample data	`database/seed.sql` (7 tables, 8 multi-item sales)	demo_guide §3	Statically parsed; live load **BLOCKED**
4	Joins, subqueries, aggregations	`database/queries.sql` — exactly 16	Run file; demo_guide §4	Count and syntax checked; live results **BLOCKED**
5	Advanced database features	`database/advanced_features.sql` (view, procedure, function, trigger)	Verification blocks; demo_guide §5	Live **BLOCKED** (integration tests exist)
6	Backend–database connectivity	`config.py`, `services/database.py`, `/health`, `python -m services.database`	demo_guide §6	Config/error handling PASSED (unit); real connection **BLOCKED**
7	CRUD and transactions	`services/*.py`, `app.py`, `templates/*`, `services/sales.py`	UI workflow; demo_guide §7–§8	Logic PASSED with fakes; Oracle transaction/locking **BLOCKED**

No criterion is claimed as live-verified.

29. Conclusion

The project delivers a normalized seven-table Oracle design, deterministic data, sixteen documented queries, four PL/SQL-side objects, a layered Flask application with an atomic, lock-ordered sales transaction, and a status-honest test suite. All checks that can run without Oracle pass; the live Oracle checks are ready and must be executed on the student's Windows machine to complete verification for the faculty review.

30. References

• Oracle Database 21c Documentation: SQL Language Reference; PL/SQL Language Reference; Database Concepts (data concurrency and consistency).

• python-oracledb documentation, <https://python-oracledb.readthedocs.io/>.

• Flask documentation, <https://flask.palletsprojects.com/>.

• OWASP Cheat Sheet Series: SQL Injection Prevention; Cross-Site Request Forgery Prevention.

• Elmasri, R. and Navathe, S., *Fundamentals of Database Systems*, Pearson — normalization chapters.